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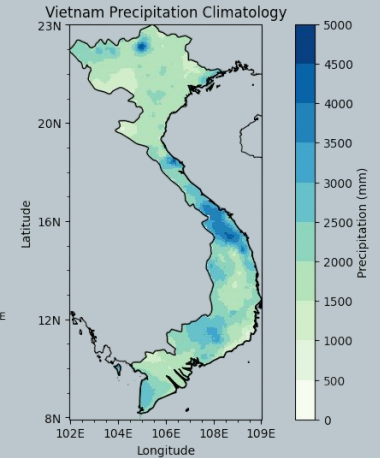
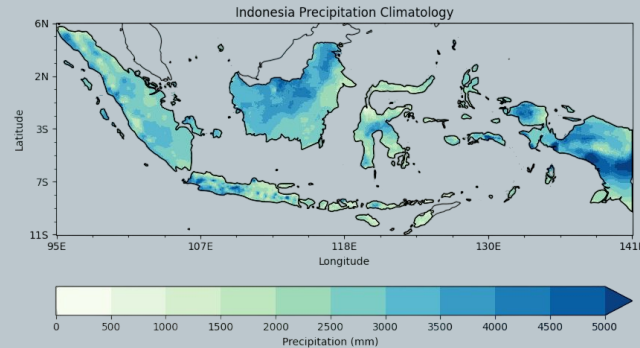
A Comparison of Drought Severity Impact on Cereal Production in Vietnam and Indonesia

GROUP CHAAC

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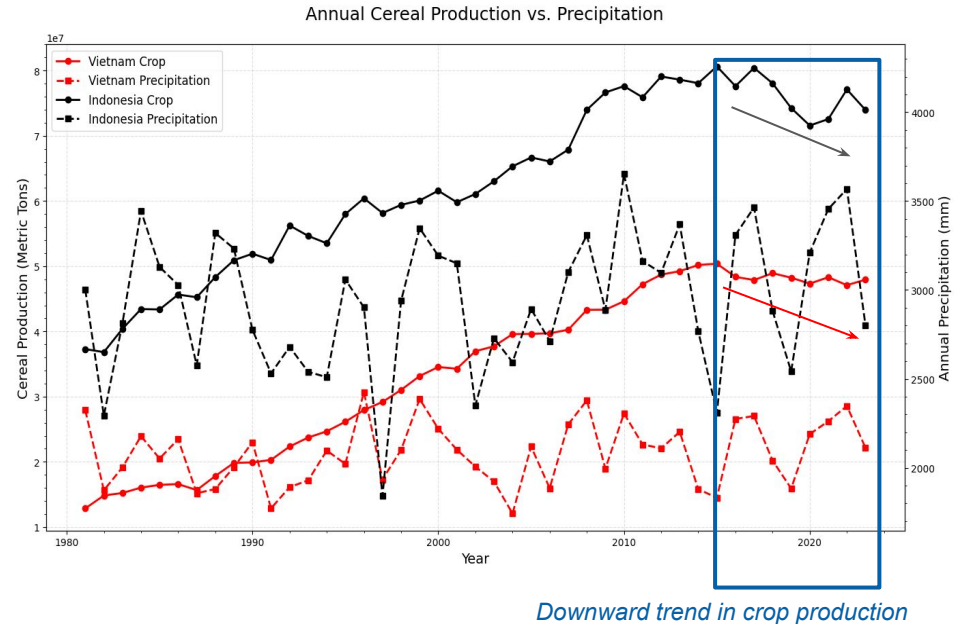
[Captions]

Jul 24, 2026

Introduction

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- Droughts and floods are becoming **more intense** in Southeast Asia (Amnuaylojaroen and Chanvichit 2024).
- Prolonged precipitation deficits (Qian 2019) may threaten **cereal production** and **food security**.
- Recent research is limited in **time** (up to 2019), **crops** (covering rice and maize only), and **sub-region focus** (impacts on mainland vs. maritime)



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- Introduction
- Objectives and Hypothesis
- Methods and Materials
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Objectives

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To assess the impacts of drought severity on crop production in Vietnam (mainland) and Indonesia (maritime) and compare the potential differences.



Hypothesis

Crop production **increases** under **mild droughts** due to higher insolation.

Crop production sharply **declines** under **severe droughts** due to extreme scorching.

The impact of drought severity on crop production is **different** between **Vietnam** and **Indonesia**.



Source: Farmvina 2026



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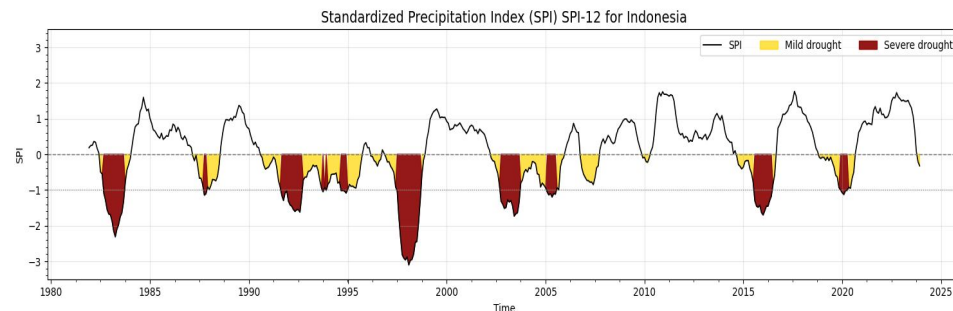
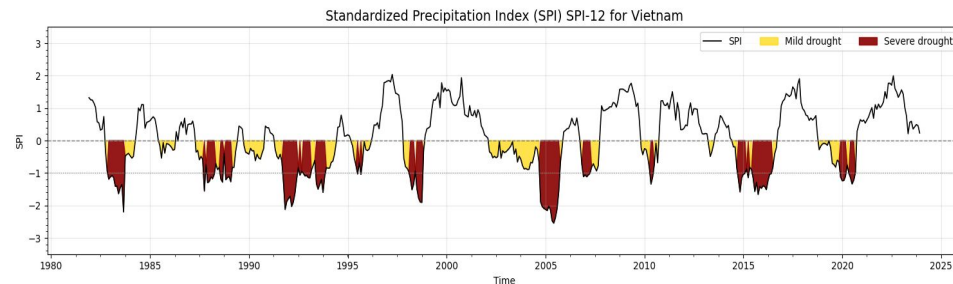
Methods and Materials

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Analysis period: 1981-2023

Variable	Data Source
Precipitation	CHIRPS v3.0 (0.25°) (Funk et al 2015)
Cereal production	FAOSTAT database (FAO 2025)

Analysis	Method
12-month Standardized Precip. Index (SPI) (Zaki et al. 2022)	12-month standardized rolling window
Drought severity (Zaki et al. 2022)	Mild: $-1 < \text{SPI} < 0$ Severe: $\text{SPI} \leq -1$
Crop production anomalies (He et al. 2026)	Linearly detrended and normalized
Pearson correlation (Wilks 2019)	SPI and crop anomalies
Principal Component Analysis (Wilks 2019)	SPI and crop anomalies



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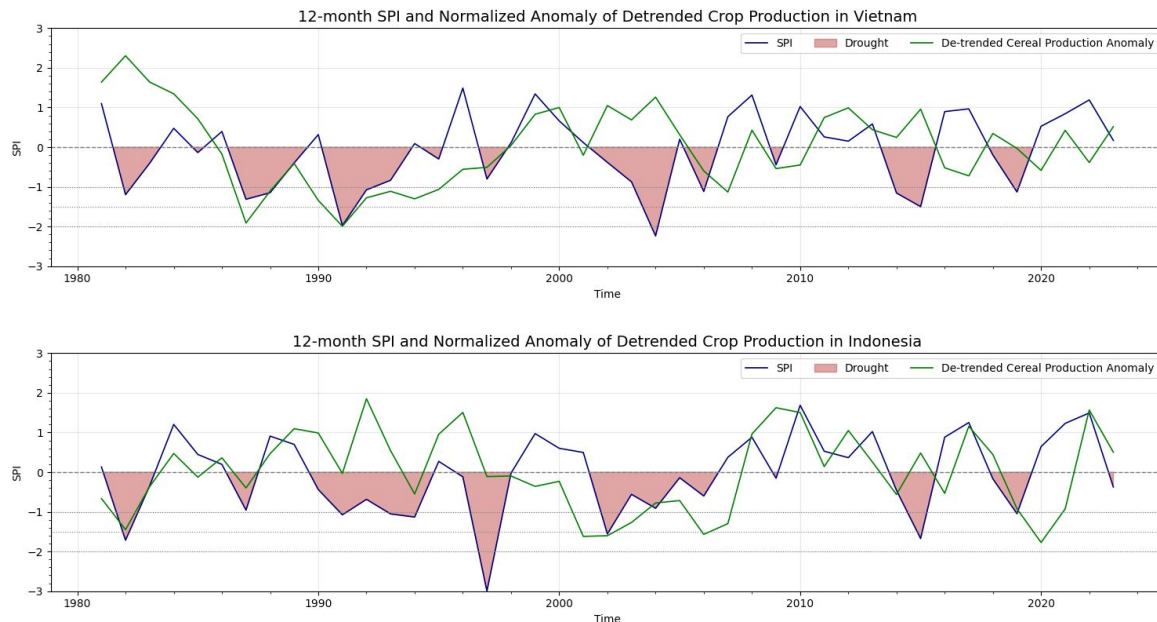
SPI & Crop Production Correlation

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We compared the annual **detrended crop production** of each country to the corresponding **12-month SPI**.

The Pearson linear correlation is **not** statistically significant.

- Vietnam: **0.10**, P-value **0.53**
- Indonesia: **0.28**, P-value **0.07**

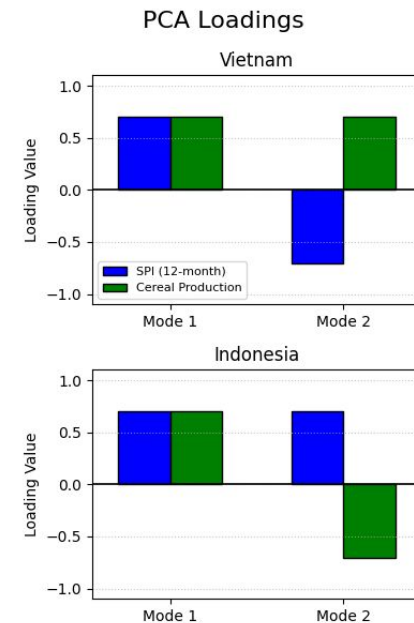
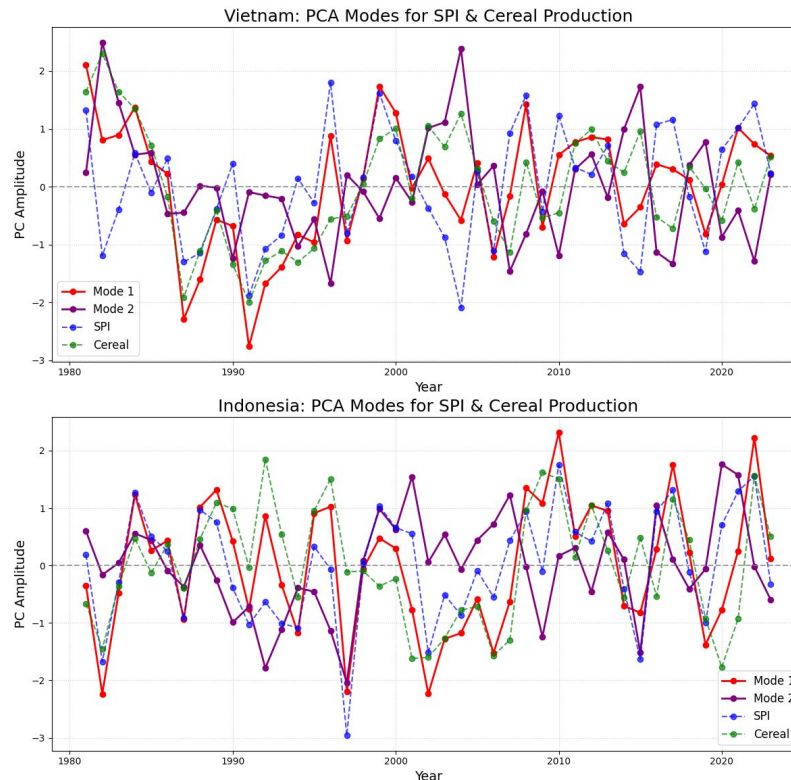


Principal Component Analysis

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- First PCA mode captures **covariant** relationship between SPI and cereal
- Second PCA mode shows **contravariant** relationship

Country	Mode	Expl. Variance
Vietnam	1	55%
Vietnam	2	45%
Indonesia	1	64%
Indonesia	2	36%

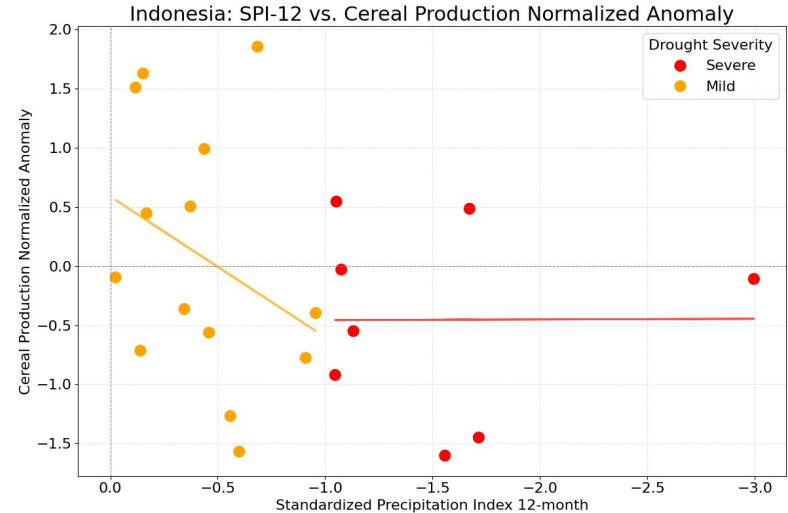
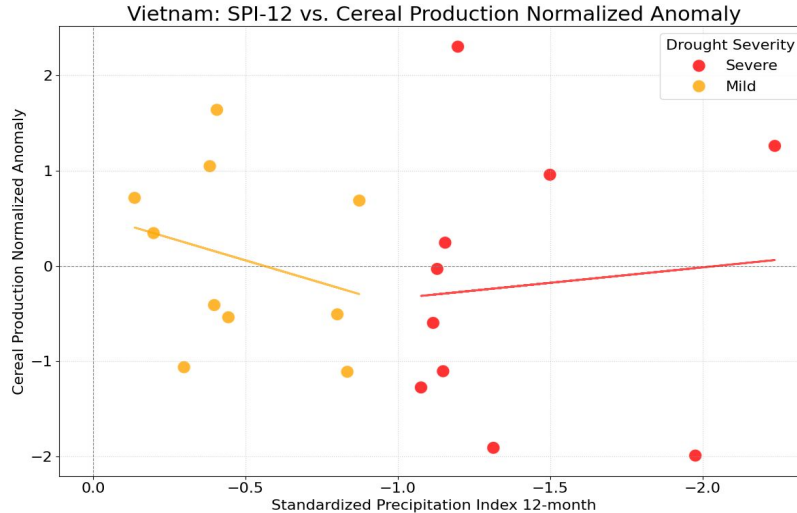


Drought Severity

- Negligible severe drought correlation
- Weak positive correlation : SPI values move to 0
- High p-values : correlation cannot be trusted as meaningful trends

Country	Severity	Correlation (r)	p-value
Vietnam	Severe	-0.092	0.800
Vietnam	Mild	0.269	0.452
Indonesia	Severe	-0.004	0.992
Indonesia	Mild	0.319	0.266

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Conclusions

The analysis **does not provide conclusive evidence** for our hypotheses of

- Mild drought increases cereal production.
- Severe drought sharply reduces cereal production.
- The drought–production relationship differs significantly between **Vietnam** and **Indonesia**.

Limitations

There are **many relevant variables** that are not yet incorporated into our analysis:

- Technological advancements in crop production
- Economic incentives and factors affecting agricultural supply chains
- Other environmental factors such as cloud cover, air quality, soil quality, etc.

Future Work

- **Controlling for more factors may further clarify** the relationship between drought and crop production, such as non-parametric detrending of crop data.



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Thank you!



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